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Digital social innovations in rural areas – process tracing and mapping critical junctures

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ABSTRACT

In recent years, digitalisation projects have helped rural areas to bridge some of the urban-rural disparities, such as accessibility issues as well as the digital literacy divide. To further exploit the potential of digitalisation for rural areas, we aim to understand the implementation of digitalisation processes and their conceptual alignment with Digital Social Innovation (DSI). As an example of Neo-Endogenous Regional Development (NERD) approaches, this perspective provides rich insights into small-scale rural adaptation in a rapidly digitising world. DSI encompasses a strong process perspective, but lacks a broader empirical application. It is a multifaceted concept that includes bottom-up participation, learning and empowerment. To operationalise the process perspective within DSI, we apply a process tracing approach within a qualitative comparative analysis of four case studies. Based on schematic mappings, we conclude with four critical junctures that provide insights for rural DSI. First, innovation can be triggered both endogenously and exogenously, with different implications for path dependencies. Second, participatory processes are crucial for digital innovation in rural communities. Third, hybridisation as an intermediate step can help to reduce the digital divide. Finally, successful institutionalisation benefits from collaborative learning. These findings outline how successful digital social innovation generates impacts beyond the project scope.

1. Introduction: rural digitalisation as an innovation trajectory?

The urban-rural digital divide has been widely discussed in rural studies. In particular, this applies to the infrastructural dimension (Park, 2017; Roberts et al., 2017). However, even if digitalisation risks bringing along potential new inequalities, it also bears potentials in empowering rural communities and valorising existing local potential. Lately, rural digital technologies and platforms are increasingly being integrated into many social practices in rural life, including remote working, online shopping, and digital health services. The COVID-19 pandemic has reinforced this trend, which contributes to compensation for (some) comparative locational disadvantages (Tim et al., 2021). Despite the infrastructural catch-up, social and generational disparities in digital skills and practices increasingly enter the debate (Moris, 2021). Conceptually, digitalisation can be understood as an overarching, exogenous impulse that meets local initial conditions and adaptations. Against this background, rural, small-scale digitalisation processes can be seen as examples of neo-endogenous regional development (NERD; Bosworth et al., 2016a; Bosworth et al., 2016b).

Within this perspective, the social innovation (SI) concept has been highlighted for valorising endogenous innovation potential, with a particular focus on local human capital and goods (Grison and Landel, 2019). A decade ago, it has been expanded to encompass a digital aspect (Kaletka and Pelka, 2015; Stojanovic, 2015). Digital social innovation (DSI) specifically describes bottom-up processes of digital innovation, with different implications for urban and rural communities (Certomà, 2020; Fahmi and Arifianto, 2022). It provides two central perspectives that frame this paper. First, beyond technical innovation, DSI also aims at changing attitudes and behaviour within a group, thus fostering new social settings (Neumeier, 2017). Second, its inherent process dimension considers both the process towards innovation and its outcomes. This bears the potential to deepen the understanding of underlying social processes, and, thus, rural innovative potentials on a more general level. Processes are to be seen as a series of decisions of (more or less)

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intentional character and that become 'visible' in form of junctures. However, this process dimension is still empirically underexamined. Against this background, we aim to understand the underlying processes of rural digitalisation. To expand this knowledge of processes within the DSI concept, we pose two research questions. First, how do critical junctures affect digitalisation processes in rural areas? Second, how do such critical junctures affect SI processes within rural communities beyond the scope of the digitalisation project? To identify the central turning points of these processes, we apply a comparative qualitative process tracing approach and operationalise these potential turning points as critical junctures (Bengtsson and Ruonavaara, 2017). Based on four cases of rural digitalisation, we derive knowledge from four critical junctures that can be identified across all case studies.

Our paper is structured as follows: Firstly, we contextualise the DSI concept in the frameworks of NERD and SI in rural regions. Based on this, we address conceptual considerations on critical junctures for our analytical framework (Section 2). We then introduce our process tracing approach and outline the visualisation of mappings (Section 3). Based on the mappings and results (Section 4), we outline some of the strands of debate identified (Section 5).

2. Conceptual framework: rural digitalisation trajectories

2.1. Neo-endogenous development and digital social innovation

For quite a while, rural studies have focused on endogenous development that valorises local potentials (Smas and Schmitt, 2014). However, new challenges and potentials from beyond the regional scope challenged the idea of an exclusively endogenous development. Therefore, neo-endogenous approaches to regional development (NERD) are becoming an increasing strand of rural development debates (Bosworth et al., 2016b). This perspective overcomes the binary between endogenous and exogenous impulses (Bosworth et al., 2016a; Ward et al., 2005). It highlights regional assets and capital while simultaneously acknowledging the role of external factors (technological advancement, geopolitics, etc.). Regarding the digitalisation context, NERD describes potentials arising from the adaptation of new technologies to local needs, capacities and perspectives (Ray, 2006). On the one hand, the mixture of endogenous and exogenous stimuli can provide new ways to valorise existing human capital and other assets of rural areas or foster cooperation with external stakeholders. On the other hand, the integration with external factors also involves new risks. Yet, empowerment is a central concern of NERD approaches aiming to enhance capacity building – even beyond a specific project scope (Navarro-Valverde et al., 2022). Empowering local communities also implicitly embedded regional development in cultural, territorial, and social terms. Embeddedness is concerned with the impact of social systems (Brinkley, 2017; Granovetter, 1985) or spatial settings on regional development decisions (territorial embeddedness; Hess, 2004). Multi-scale embeddedness picks up the spatial dimensions of NERD (Salder and Bryson, 2019).

Against this background, social innovation (SI) aligns with NERD approaches in terms of the bottom-up perspective (Bosworth et al., 2016a, 2020; Navarro-Valverde et al., 2022). It focuses on the innovation of products, practices and community networks. The debate often links innovation and creativity to urban phenomena such as agglomeration advantage (Florida, 2008). Yet, more recent contributions have expanded the focus on rural regions (Eder, 2018; Glückler et al., 2023; Rösch, 2000) and the potential of traditional knowledge and community networks (Sorensen, 2009; Spielman et al., 2011). Hence, in embedding innovation in these local capacities, SI resumes a people-centred perspective on innovation (Neumeier, 2012, 2017; Novak et al., 2018). Thereby it promotes new practices, networks, outcomes, organisational approaches, and the maximisation of local resources (Bosworth et al., 2016a).

Recently, the rise of digitalisation has brought about a variant of the SI concept: digital social innovation. DSI describes innovation processes with a narrower focus on digitalisation in rural areas and its underlying potentials for social innovation (Ozman and Gossart, 2018; Rodrigo et al., 2019). This refers to the specific solutions for rural development and its inherent demographic challenges induced by digitalisation (Rundel and Salemink, 2021; Salemink et al., 2017).

Empirical studies on DSI illustrate that innovation goes beyond technological implementation, including digital literacy and process innovation (Bria et al., 2014; Rodrigo et al., 2019). This underlines that not every process of digital implementation must necessarily lead to (digital) social innovation. Rather DSI focuses on the process of and the innovative approaches involved: bottom-up participation, collaborative learning and empowerment (Sept, 2020; Zerrer and Sept, 2020).

Bottom-up participation: Firstly, the actors involved in digitalisation are central to the innovation process (Zerrer and Sept, 2020). This concerns both endogenous and exogenous stakeholders and communities involved. Their internal interaction and the resulting governance are central to understanding the underlying process (Georgios and Barraí, 2023). A focus on the actors also allows to scrutinise the local impact of such processes. Accordingly, joint experience in DSI processes can raise community identification, and place attachment in the end for the involved and addressed people (Birnbaum et al., 2021).

Collaborative learning: Second, the focus on the digital divide is increasingly shifting from infrastructural concerns to more social barriers regarding acceptance and skills (see OECD, 2023; Williams et al., 2016). Identifying innovative ways to adapt digitalisation processes can foster local collaborative learning and knowledge co-creation in the short term, and raise acceptance for innovative paths in the long run (Novak et al., 2018; Rundel and Salemink, 2021). This social process entails employing and exchanging skills within their communities. The resulting learning processes include skill enhancement but also innovative approaches to (joint) problem identification, participation, partnership organisation, and cooperation (Labianca, 2021; Navarro-Valverde et al., 2022). Similar ideas have been referred to as communities of practice in other disciplines with a stronger focus on the social dimensions (Pyrko et al., 2017).

Empowerment: Finally, empowerment considers the long-term learning impact in social innovation. Thereby, DSI expands beyond the technological outcomes and derives more detail on the processes themselves. This empowerment refers to both community and institutions (Certomà and Corsini, 2021; Rumbul, 2016). It involves taking control over skills, data, and new technologies by promoting the use of collective knowledge and bottom-up initiatives (Certomà, 2020). A key element of this perspective consists in open, collaborative and low-threshold approaches which can be seen as opposed to economic competition. Open-source and collaborative digital applications are therefore relevant examples of DSI (Cangiano et al., 2017).

2.2. Critical junctures: change and continuity

The critical junctures approach provides an analytical lens for understanding transitional decision points. Critical junctures have the potential to reshape outcomes and open up new trajectories (Bengtsson and Ruonavaara, 2017; Hogan, 2006; Soifer, 2012). They often mark certain historical moments and might have long-term and transformative effects (Soifer, 2012). Critical junctures are thus such moments that have the potential to bring about change within institutions and beyond (Cortell and Peterson, 1999), but not necessarily in a negative way. Soifer (2012) elaborated on potential conditions that can trigger critical junctures. While *permissive conditions* include the underlying contextual conditions like agency or power structures, *productive conditions* refer to structures that mark the end of the juncture. Empirically, these junctures can thus be triggered by long-term developments or short-term disruptions (Hogan, 2006; Liu et al., 2014; Mahoney, 2001).

Critical junctures received great attention in institutional change theory (Greif and Laitin, 2004; Lin, 1990) and innovation theory (Glückler et al., 2023). They describe trajectories that deviate from the

expected path. On a general level, this includes the re-organisation of networks, new regulatory frameworks, a shift of political power, or of organisational structures, while on the small scale, this might involve the introduction of innovative tools and approaches to local issues, as outlined by the SI literature. Institutional change refers to institutional implications of (technological) innovation and resulting changes in institutions. Empirical examples of digitalisation as a trigger of institutional change are broadly available (e.g., Budzinski and Stöhr, 2019; Fahmi and Arifianto, 2022). By contrast, continuity is conceptually linked to the idea of path dependencies. In economic geography, path dependencies describe situations where earlier decisions significantly influence subsequent events and outcomes (Bergek and Onufrey, 2014; Sewell, 2011). Prior sequences of "interrelated events" (Sydow et al., 2018, p. 159) affect such junctures and impede innovation (Martin, 2010). Hence, path dependencies typically refer to past events and conditions that regulate decision-making and result in lock-ins (Aarset and Jakobsen, 2015). Thus, path dependences can result in the continuation of current trajectories despite the emergence of opportunities for change (Capoccia and Kelemen, 2007). Empirically, the path dependency perspective offers useful perspectives for understanding hampered innovation processes in hindsight (Hassink, 2005; Isaksen and Trippl, 2017). With regard to SI, path dependencies are documented prior to the initiation of new approaches (Misuraca, 2019; Moulaert et al., 2005), during the institutionalisation of innovative approaches (Aarset and Jakobsen, 2015). In particular at the initaition stage of innovative approaches, path dependencies can become a major constraint to rural innovation. This highlights both the policy as well as the theoretical need to apply the path dependence concept on rural DSI. While there are few contributions in the field of SI (Richter and Christmann, 2023; Steiner et al., 2023), the DSI literature lacks a systematic understanding of the underlying processes beyond single case studies. We therefore aim to develop a conceptualisation on rural digitalisation processes both in terms of their procedural composition (critical junctures, path dependencies) as well as the spatial dimension of impulses (NERD).

3. Data and methods

3.1. Case studies

Our analysis draws on a Europe-wide screening of projects that deal with rural development and approaches this topic with the help of the potential of digitalisation. From a database of 193 projects of rural digitalisation projects (Birnbaum et al., 2021), a filtering process based on a variety of criteria concluded in four cases. The cases bear similarities in their (a) a content focus on place attachment, (b) current or recent activity at the time of analysis, and (c) identifiable key actors and contact persons. On the other hand, they complement each other in (a) project constellations, (b) resource availability, (c) funding frameworks, and (d) geographical balance. The filtering process resulted in two streams of rural digitalisation processes, applicable to the majority of these projects. Two of the projects aim to promote digitalisation as such. These are represented by the case study examples *Bremke.digital* and *Digitales Dorf Spiegelau-Frauenau*. The two seek to increase place attachment via digital tools illustrated by the projects *Nestbauzentrale* (Germany) and the Campaign *Get the MOST* (Austria, see Fig. 1).

To increase the validity of the qualitative approach, the comparative case study design with two cases per process approach aims at systematising this field (Bartlett and Vavrus, 2017; Boddy, 2016). While two cases each are certainly not representative for rural digitalisation as such, their good practice character makes them exemplary for these two streams of approaches. The good practice refers to the successful institutionalisation beyond the project scope, as well as broad knowledge on success factors and barriers.

3.1.1. Case A: Bremke.digital

Bremke.digital is an example of a local communication and exchange platform, developed by a cooperation partner and digitalisation experts as to the inhabitants' demands. A set of apps to support local networks and social exchange were the most prominent products of the citizen-centred approach. Initially, a newspaper article on a project idea of the House of Church Services of the Evangelical Lutheran Regional Church of Hanover triggered local politicians in Bremke, a municipality with under 800 inhabitants in the district of Göttingen. The mayor took this article as an impetus to apply for a local model project within the framework of the Land.digital funding guideline from the Federal Ministry of Food and Agriculture, through which the project was funded from 2018 to 2021. The cooperation partner for the technical implementation was the Fraunhofer Institute for Experimental Software Engineering (IESE). The group interviews were conducted with two representatives of the cooperation partner, the principal local administratetive stakeholder, and four members of the voluntary steering committee.

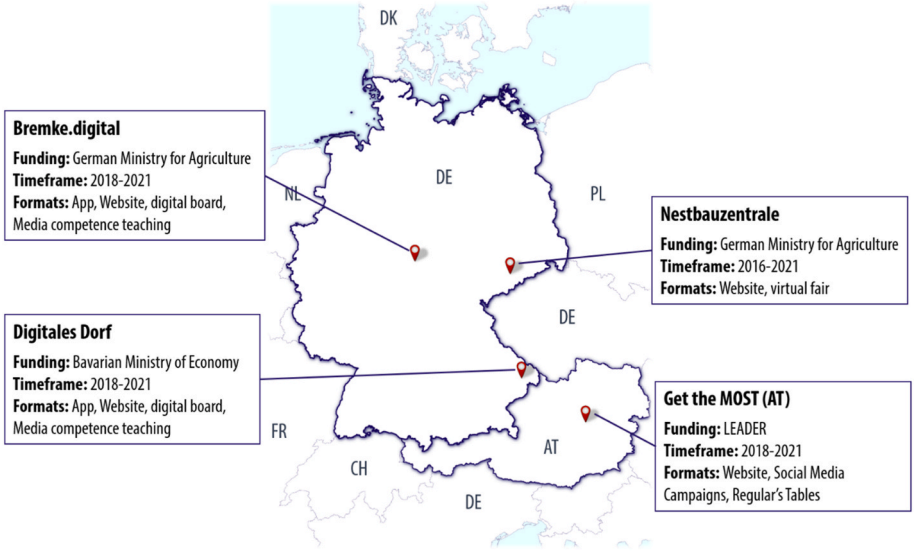


Fig. 1. Overview of Case Studies (own visualisation, based on Federal Agency for Catography and Geodesy, 2013.

3.1.2. Case B: Digitales Dorf Spiegelau-Frauenau

The project aimed at providing digital tools and solutions for improving the community life. The two municipalities of Spiegelau and Frauenau jointly took part in the Digital Village Bavaria competition, even across administrative boundaries. The project improved municipal processes through digitalisation and to attract new citizens. An important element of the project is skills enhancement, which is realised, for example, in the form of meeting points for senior citizens where volunteers give workshops on how to use digital devices. The Technical University of Deggendorf both initiated this competition and provided scientific as well as technical support in developing several apps (e.g., for the town hall, schools, sports club). All digital offers link back to the central website *dahoamviernull.de*. Several citizen participation formats contributed to process and product specifications. The group interviews included one administrative representative per municipality, a project manager of the cooperation partner and one volunteer.

3.1.3. Case C: Get the MOST

The online campaign "Get the MOST" aims to attract potential returnees to the Austrian "Mostviertel" region (Lower Austria). The starting point of the campaign was a demographic analysis of the LEADER region Eisenstraβe. The project "youngCitizen Relationship Management" (2018–2020) was implemented as a central element within the field "Demography and Young People" of the Local Development Strategy, with the campaign being the central element. It aims to address young academic and professional graduates to inspire them to return to the region. The campaign uses a mixture of digital storytelling and personal contact points (returnees' regulars' tables, excursions), and attempts to highlight the region's special features that are rather marginal in the public perception. The group interview included a political representative as well as the two project members of the project team.

3.1.4. Case D: Nestbauzentrale

Nestbauzentrale (translate 'nesting centre') is a rural demography project in Saxony (Germany). It aims to provide an information platform and contact point for newcomers and returnees. Building on a demographic strategy and a regional future concept, the district of Central Saxony developed the idea of a return agency. The aim is to disseminate information for families moving into the district and to integrate them locally (jobs, childcare). It links the demographic dimension with the regional construction sector. Relevant formats include websites, a personal contact point, a virtual fair (e.g., a rural construction), and cross-channel campaigns. The group interviews included two administrative representatives, the former project manager from the cooperation partner as well as two representatives of regional development (LEADER).

3.2. Qualitative process tracing

The process tracing approach, as originally introduced by George and Bennett (2005), refers to the identification of causal chains and mechanisms that link a sequence of events. In its original conception, it was based on statistical testing of specific variables and the systematic examination of historical documents (George and Bennett, 2005). In the present study, we employ a qualitative approach to apply a comparative analysis to understand processes in depth (Bengtsson and Ruonavaara, 2017). Qualitative process tracing focuses on specific details of the case to uncover important factors that contribute to the actual outcomes of the respective process. Therefore, these process outcomes (i.e., a product, a specific law, or institutional change) are the starting points of analysis. The approach then entails a chronologically backwards analysis of events that led to the outcome (Harvey and Kou, 2013; Trampusch and Palier, 2016).

Due to the comparative approach, it is possible to identify plausibly transferable patterns and variations across multiple cases of a similar phenomenon (Bartlett and Vavrus, 2017). Crucially, periodisation provides a comparative systematic of the sequentiality of events (Harvey and Kou, 2013). It allows for an examination of the role of certain events and their linkages between these events, with particular regard to their impact on the process. These can include (a) political focal points of institutional intervention, (b) social mechanisms of interaction leading to cooperation or conflict, (c) the context of norms, structures or resources, as well as (d) critical junctures (George and Bennett, 2005). The comparative perspective provides insights into such transitional situations beyond the specific context (Bengtsson and Ruonavaara, 2017).

3.3. Data collection and analysis

We draw on a comprehensive approach that combines document analysis and group interviews. The comparison of the four rural projects as introduced in Section 3.1 projects served as the basis for our analysis.

Document review involved the examination of relevant documents related to the digitalisation initiatives. These included publicly available reports (e.g. LEADER evaluation, annual reports), project plans (e.g. by funding bodies, project fact sheets), event documentation and other official records that provide insight into the development and implementation of the case studies. Relevant dates (events, project milestones, etc.) were distilled and condensed regarding their key elements (involved actors, decisions, involved resources). These dates were the basis for group interviews. Interviews with key project stakeholders included policymakers, engaged members of the public and project managers (for interviewed stakeholders by case, see Section 3.1). Based on the events and dates from the document analysis and an interview guide, group interviews of two hours each were conducted. Through guided group interviews, we identified gaps in process understanding and outlined decision-making processes. Four aspects were central during the interviews: points of time, where relevant decisions were made in terms of content and formats, who was (not) involved and which resources (human and financial) were available during the decision. Each group interview resulted in a large-format graphic documentation of timelines, transcripts and audio documentation. The condensation of the results relied on the graphic output of the workshop and the notes, while the audio files were used for validation. The condensation involved three central components:

Firstly, a periodisation of the individual projects was based on the cross-case comparative analysis (Bengtsson and Ruonavaara, 2017; Harvey and Kou, 2013). The step involved identifying and coding stages in the case study to structure and visualise the sequence of events. This is in line with operationalisation of critical junctures in other studies (Bengtsson and Ruonavaara, 2017; Sorensen, 2009). In doing so, we applied an inductive approach as we developed a set of project phases that are applicable across all four case studies. Also the qualification of critical junctures by cross-case comparison follows the inductive logic. At the same time, the inductive approach is built on general categories reflected in literature on project development, e.g., the categorisation of process phases (Christmann, 2020). These elements might be regarded as decuctive elements, but our operationalisation is of mainly indcuctive character (Table 1).

Table 1
Operationalisation of the critical junctures approach, based on Bengtsson and Ruonavaara (2017). The Numbers given locate respective junctures within the mappings.

①	Initial Momentum	How were digital projects initialised?
②	Objective Formulation	How was the strategic orientation set?
③	Process definition	In which formats did the process definition result?
④	Institutionalisation	What was the decision process for long-term adoption?

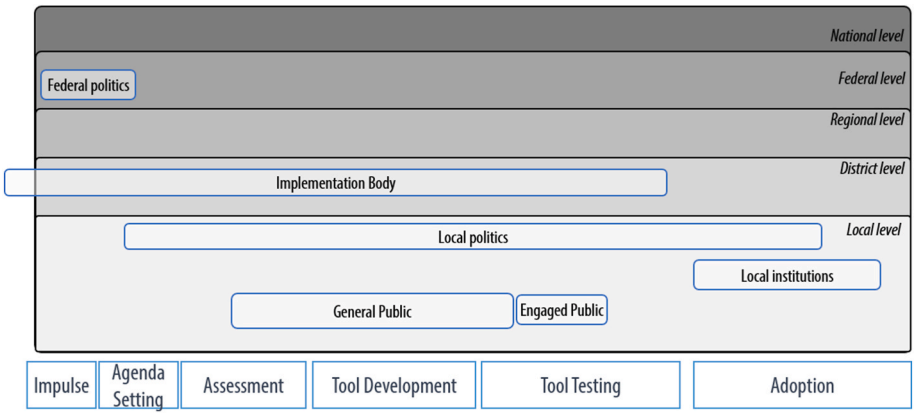


Fig. 2. Mapping Schematic.

Secondly, a schematic mapping provides an overview of the actors and institutions involved in the processes (for other applications see for example Wilhelm, 2023, see Fig. 2). This helps to identify the different actors and their roles in shaping initiatives (Campbell and Gregor, 2002). Following other institutional mapping approaches (Chilla et al., 2012, Chilla and Lambracht, 2023), it applies a multi-level reference, represented by grey background layers. They range from the regional level to the global level from the inside out. In the following, we refer to regional as the subnational federal state level.

Third, following the logic of network analytical approaches, we mapped the interactions of stakeholders throughout the process. Along with Schneider and Rohlfing (2013), the extraction of interactions between stakeholders and their roles throughout the process provides relevant insight into the causality of processes. This helps understand how collaborations, conflicts, and negotiations among actors shape decision-making. In this application, the interactions of stakeholders are mapped concerning their interaction in specific decisions (e.g., format or content) and their spatial context.

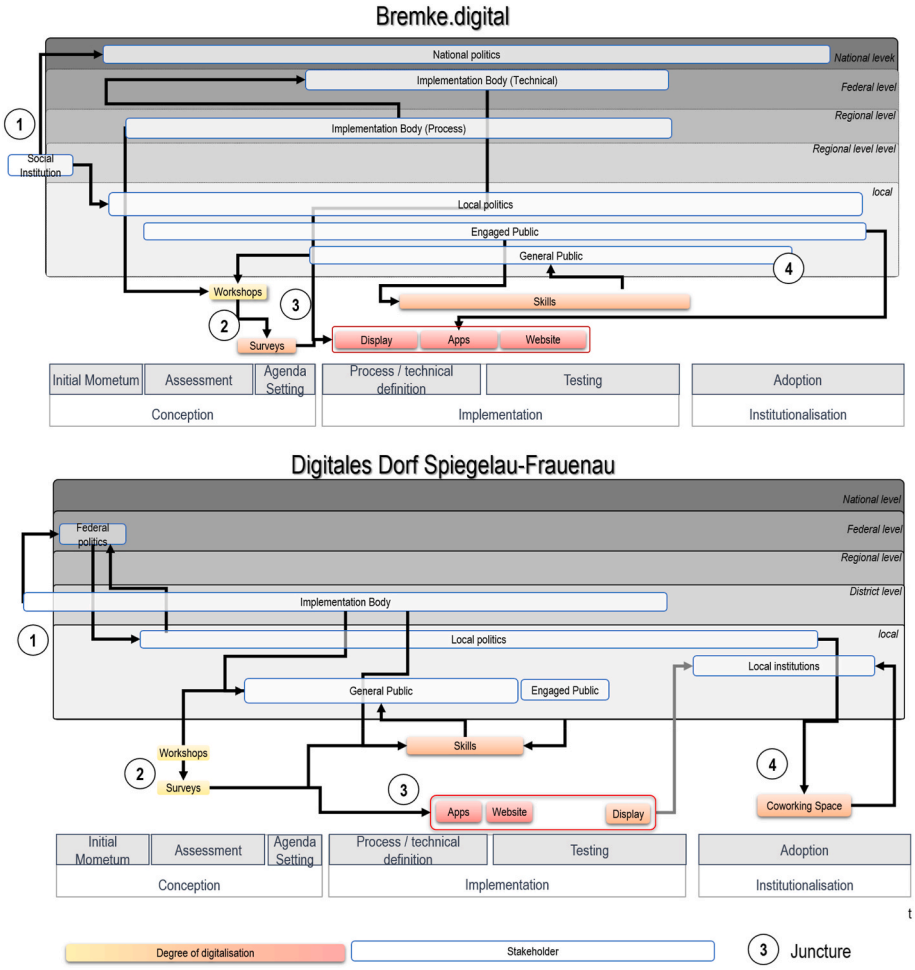


Fig. 3. Mappings of bremke.digital and Digitales Dorf in a comparative presentation.

4. Results

4.1. The digitalisation approach

Schematic visualisations illustrate the project development processes of all four case studies. Due to strong similarities among the sets of projects, the visualisations are merged in pairs: *Bremke.digital* and *Digitales Dorf Spiegelau-Frauenau* on the one hand (Fig. 3), and *get the Most* and *Nestbauzentrale* on the other hand (Fig. 4). They represent an abstraction of each of the digitalisation processes and visualise relevant stakeholders across scales (vertical) and process phases (horizontal). The arrows represent activities and interactions between stakeholders that led to the project development.

Bremke.digital (Case A) and Digitales Dorf Spiegelau-Frauenau (Case B) aimed to develop comprehensive digitalisation projects with a focus on the local level. In both cases, regionally embedded institutions initiated a local process of developing strategies to promote rural quality of life. This initial momentum, marked as (1) in Fig. 3, consisted of a newspaper article (Case A, church) and the demands of a nearby university campus (Case B, scientific institutions), respectively, that both involved the idea of bottom-up processes for the respective rural communities:

> "The digital world should not be imposed on us. Instead, we need to think about how digitalisation can support the things we need and want in our village." (Case A)

The rather broadly defined aim was the basis of the incorporation of digital tools at an early stage of project development. In both cases, the political stakeholders applied for exogenous pilot project funding at the national (Case A) and federal level (Case B), respectively. In the latter case, the implementing body and the Bavarian Ministry of Economic Affairs jointly developed a new funding instrument for model projects. Moreover, in both cases, the respective implementation bodies were chosen based on a combination of local embeddedness and technical expertise in digitalisation. The actors' local embeddedness moreover promoted public participation processes:

> "Then people come up from somewhere in the region and say, I'm interested in digital topics and want to get involved, I'll look through the images once a month, or I'm the person who can support technically. We connect and then things work out."(Case A)

Moreover, both cases build on extensive public consultation and iterative participation processes, which resulted in community-based objective formulation (2). In Case A there was a joint demand for communication and cooperation tools that should strengthen social bonds. Meanwhile, residents of Case B stressed the importance of interrelated sectoral demands and therefore requested digital solutions that simplify and integrate these fields of public life (e.g., mobility, healthcare, civil engagement):

> "We wanted to pursue the idea of simply putting rural areas on a higher standing compared to urban regions through digitalisation. That was our idea, our thought, our concern. In the process, we realised that this cannot be achieved in a single topic or in a single

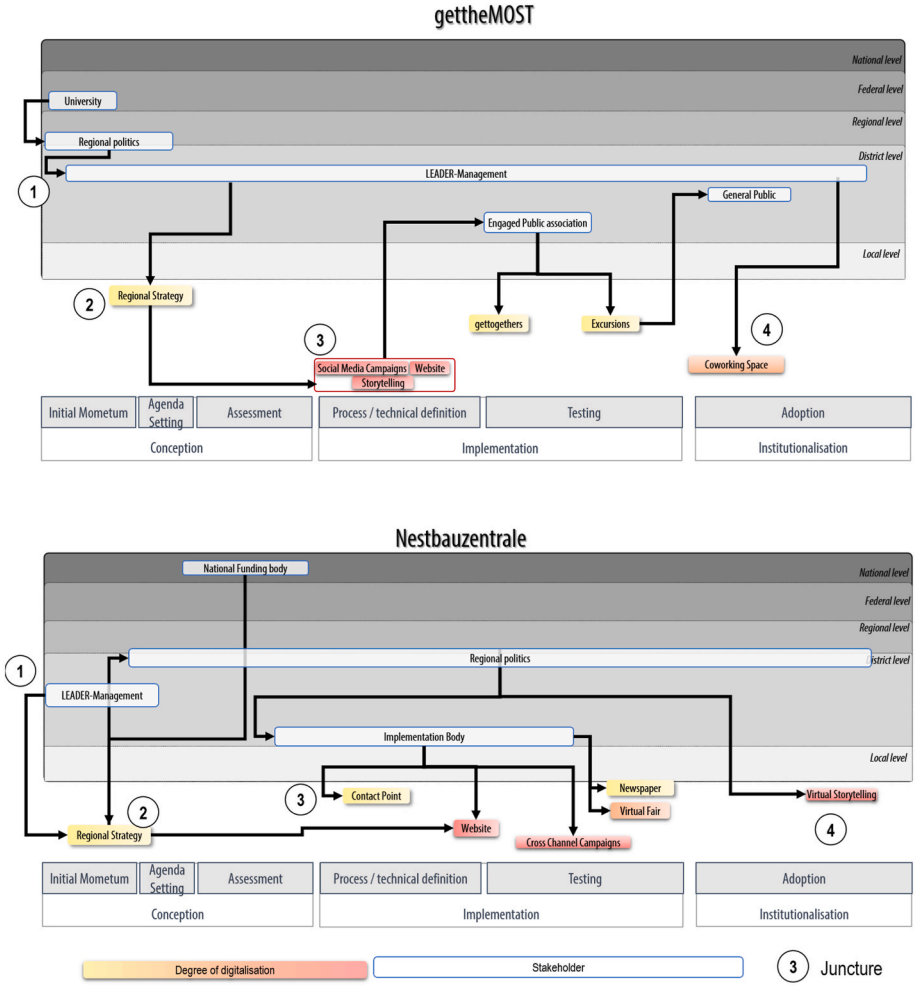


Fig. 4. Mappings of bremke.digital and Digitales Dorf in a comparative presentation.

field of action. We then defined a bouquet of topics in which digitalisation can have an impact in rural areas." (Case B).

Process definition (3) likewise involved public participation. Both cases experimented with different stages of (non)digital tools and focused on an assessment of skills and capacities. A central result in both cases was the demand for offers that would enhance digital skills. Learning was therefore a key element of the process roll-out: The emergence of learning opportunities in the sense of a community of practice has enabled a larger proportion of citizens to participate in the products developed by the community of practice. Finally, this resulted in similar products. These included a range of digital tools (websites, smartphone applications) and hybrid formats (skills training, digital boards). The argument for hybrid formats was the aim of making digitalisation "tangible" and lowering thresholds to increase acceptance. As an example of the learning focus, the implementing body (Case B) responded to a specific demand of the public, by developing a "swiping app" to introduce the logic and use of mobile apps. Moreover, regular tables for such skills training networked different skills within the communities of both cases:

> "I joined via the digital regulars' table that developed from the group of people who took part in the events. There were various events on how to use a smartphone or how to edit images." (Case B)

In both cases, the tools succeeded in institutionalisation (5). In both cases, local stakeholders, provide this long-term maintenance. Case A established a citizen committee that monitors the tools. This evolved as a central success factor and enabled the implementing body to roll out this format:

> "We had already written the transfer into our application, but we didn't expect it to work out so well. We were able to utilise the coronavirus pandemic when we had a calculation model - what does it cost for a village, what for the municipality? And this has shown that there is interest throughout the community." (Case A)

By contrast in case B, appropriate local organisations (churches, schools, …) adopted maintenance, even though this led to the decision to turn off some tools in the case of Digitales Dorf Spiegelau-Frauenau.

4.2. The process approach

Get the MOST (Case C) and *Nestbauzentrale* (Case D) resulted from a regional decision for a demographic project by LEADER-funded entities (1, see Fig. 4).:

> "The municipalities would never have been able to finance this. That's the good thing about the LEADER approach, you can experiment without risk." (Case C).

However, in the case of Get the MOST, exogenous university experts were assigned by local politics to examine the status quo, whereas, in the case of Nestbauzentrale, a national funding scheme triggered regional actors to apply.

In both cases, an ex-ante agenda was set. The objective definition involved the responsible political bodies but also external consultation (2). The result was the issue of demographic change as the basis of the funding application.

> "A development strategy was drawn up long beforehand. The LEADER managers were among those involved. They were asked what was missing - and the result was: emigration and a lack of labour. That's when the project idea was born." (Case D)

Public consultation did not play a role in the process definition. Rather target economic sectors were included punctually.

A key result from process definition (3) were measures to target demographic change, whereas digitalisation was not a central issue. On this basis, both digital and non-digital tools were developed:

> "Digitalisation was never the goal here. It started with radio/ TV clips and people who went to trade fairs and represented the region." (Case D)

For instance, physical contact points (get-togethers, excursions, and newspaper marketing) assured low-threshold entry points, while a website and cross-channel marketing are the digital backbone of the projects. In the case of Nestbauzentrale, a virtual fair on construction and living complements the digital tools. In this case, there was innovation in both technology and social practice. This arose from the COVID-19 pandemic when stakeholders sought to digitally adapt an established format to meet curfews and meeting restrictions. This relied heavily on the learning capacity of the organisers (public administration, craftsmen) and the target groups (young people). Although the experiment was successful and led to other virtual applications, in both projects the need to provide physical points of contact was a central argument in format decisions.

> "It was hard work to get them to participate digitally. The best experience was the feedback afterwards: 'Oh, I get my contacts digitally too!' Nevertheless, we still have to see how it develops. One said, digitalisation can complement to get access information, but it can never replace personal exchange." (Case D)

Long-term adoption (4) was secured through additional financial resources by the municipalities. In the case of Get the MOST, a follow-up project allowed the foundation of a coworking space, whereas the administrative body of Nestbauzentrale determined a stable budget for long-term structures. In both cases, new ideas were implemented shortly after the project ended, that involved higher degrees of digitalisation, including storytelling or coworking.

> "We have now been on the market for three years. Before that, it was a development and now the format has become established. We have also involved the public. The feedback was great, they said we'll stick with it, we're happy with it." (Case C)

5. Discussion

5.1. Critical juncture 1 – initial momentum: neo-endogenous project initialisation can overcome the risk of path dependencies

Our findings illustrate different elements of the DSI concept and its linkages with the idea of NERD. Fig. 5 visualises the chronological structure of the projects and the empirically deduced options within the project study scope.

At the outset of a project, the initial momentum plays a crucial role as it has a cascading effect on the rest of the project stages. Based on the origin of the initial impulse, we differentiate if it was either of more endogenous (Cases A, B, C) or exogenous character (Case D). Exogenous factors such as funding and digitalisation trends are a central premise of all projects. Moreover, they often are significant drivers for the development of rural regions but have to be reflected in a differentiated way (Salamonsen, 2015; Trippl et al., 2018). First, external trends and technologies usually suffer from high entrance barriers, such as the digital divide (Morgan, 2013; OECD, 2023). Second, exogenous factors may not be flexible enough to account for local specificities when funding eligibility depends on pre-defined programme frameworks (Dawley, 2014). These often inflexible frameworks can meet public administration staff that is used to strong sectoral thinking, which again can hamper the understanding of digitalisation as a cross-cutting issue. This reflects previous studies in defining the inituiation stages of rural innovation as rather vulnerable to path dependencies (Misuraca, 2019; Moulaert et al., 2005). In contrast, an endogenous impetus involves broader stakeholder participation and debate before the funding application, which is characterised by more integrated and experimental perspectives (Glückler, 2014). On the other hand, this approach often

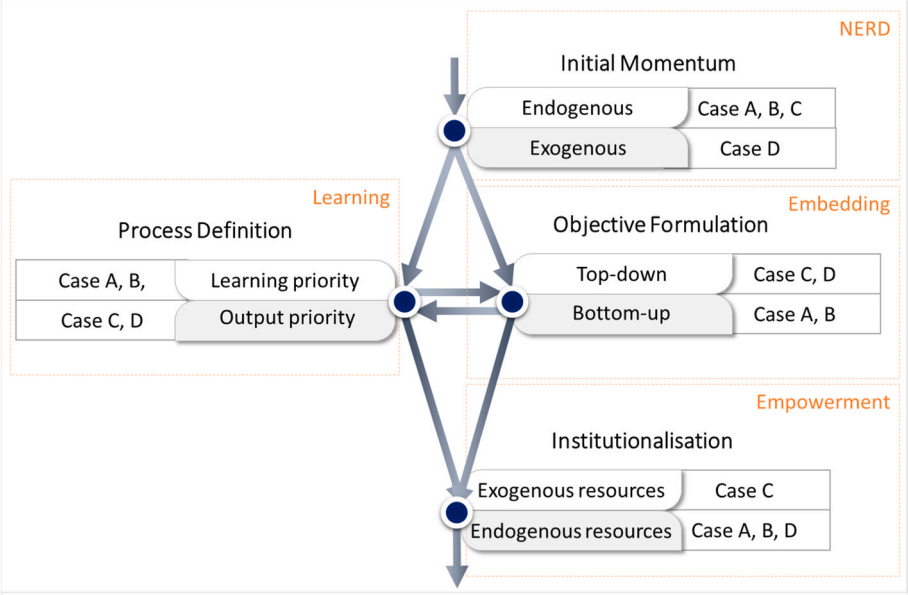


Fig. 5. Critical junctures within the processes.

comes along with limited transferability and is not always very efficient.

When exogenous dynamics meet rural regions, path dependencies can play an important role. In particular, the sectoral logic and local entry barriers can be explained as path dependencies, hampering innovation (Gibson, 2016). The combination of endogenous and exogenous impetus bridges their respective path dependency risks by combining their strengths. The digitalisation trend is therefore a strong empirical example of the NERD perspective, as it requires local adoption. While endogenous triggers can succeed in stimulating development processes at the local basis as such, exogenous funding and trends like digitalisation are indispensable drivers (Eder, 2018). A combination of endogenous and exogenous factors can therefore be seen as complementary facets of rural local development. Both can facilitate innovation, especially when these leverage local potentials and capacities (Belliggiano et al., 2020; Isaksen and Trippl, 2016, 2017). Our comparative case study tends to support the argument that rural regions benefit from a combination of endogenous and exogenous stimuli towards innovative processes to overcome path dependencies.

5.2. Critical juncture 2 – objective formulation: embeddedness of rural innovation

Objective formulation has been a critical juncture in all presented cases. The underlying processes of objective formulation can be differentiated with regard to public participation opportunities. We find predominantly top-down processes (Cases C, D) and bottom-up objective formulation (Cases A, B). Our study shows that top-down objective formulation by public authorities tends to focus more on internal consultation processes (e.g., inter-departmental exchange) than on the involvement of a broader public. A lack of flexibility to community suggestions can be a barrier to process innovation. Resulting path dependencies run the risk of exacerbating the digital divide, by underestimating the learning potential and human capital.

By contrast, various formats of bottom-up participation (surveys, events, consultation) involve a higher sensitivity to local human capital in objective formulation (Banerjee et al., 2023). The bottom-up cases were more responsive to competing interests and took an integrated perspective. Empirically, the participation phase facilitated an innovation orientation and helped to avoid path dependencies. At the same time, a lack in participation or alternative wishs uttered could have stalled this process by extending the need fo debate. This confirms a central argument of the SI perspective, which illustrates the collaborative development of innovation among diverse groups as a facilitator for comprehensive and innovative solutions (Neumeier, 2017).

Furthermore, a successful participation process can foster local embeddedness of innovation. The idea mainly stems from regional economic geography, in particular as it concerns technological innovation. However, (digital) social innovation has received rather little attention (Kompella, 2019; Terstriep and Rehfeld, 2020). However, our results show that the embeddedness of innovation in local structures through participation can increase acceptance. This is because co-creation can be considered as a strategy for embedding digital technologies within rural communities. Joint problem identification and the search for solutions under the participation of various rural stakeholder groups enhance local benefit and reflect local context (Gertler et al., 2000; Rutten and Boekema, 2007; Tregear and Cooper, 2016). This also increases the social and territorial embeddedness of the resulting structures (Hess, 2006) and might be a potential for the institutionalisation of technological innovation (Fløysand and Jakobsen, 2008).

5.3. Critical juncture 3 – process definition: hybridisation and learning

The process definition can be categorised as either learning priority (Cases A, B) or output priority (Cases C, D). Although both paths led to digital projects in the end, they followed different goals. Within the *learning priority* projects, both technological product and process innovation were the primary aim. This led to a variety of digital formats but also requested skills enhancement on the user side. Thus, in the long run, it involved both community-based technical learning as well as new tools within the administration. This indirectly brought process innovation into focus (participation formats, skills enhancement). A learning environment that combines local assets (skills) and needs (lack of digital literacy) can be considered a key dimension of process innovation. A central element of this learning element consisted of hybrid tools (e.g., digital boards). These were implemented to make digitalisation tangible, thus increasing the public's acceptance.

The projects with an *output priority* aimed at specific sectoral outputs, namely demographic development. Even though these projects addressed high-skilled target groups with supposedly high innovation potential, the original project plans left little room for learning. Product innovations (such as virtual fair, coworking space) can be seen as a by-product rather than a tool to leverage the process' aims. Even if learning

was not a priority as such, it took place. Administration staff and implementing bodies benefitted from learning processes, which built the basis for technological innovation and new organisational structures (Neumeier, 2012; Novak et al., 2018).

This juncture illustrates two issues regarding the learning dimension within DSI. First, product and process innovation are closely interlinked. DSI must not necessarily be the key aspect of projects to unfold innovative features. Even though external factors can be an additional trigger (Scheidgen et al., 2021), the willingness to think together and to learn is a key resource (Pyrko et al., 2017). This requires an in-depth understanding of local resources and respective capacities (Cowie et al., 2020; Kenny and Regan, 2021). Our case studies underline that this is linked to collaborative learning: collective learning processes contribute to joint problem identification and common knowledge on available endogenous human resources (D'Ambrosio and Boriati, 2023; Ozman and Gossart, 2018).

Second, particularly when the public utters the need for "slower" digitalisation paths, learning is an essential element for acceptance. This is particularly true considering the digital divide as a 'literacy divide' (Morris et al., 2022). Public places in the 'analogue' world that allow for getting in touch with digital and hybrid tools can help alter the willingness to learn (Kaletka and Pelka, 2015). This kind of capacity building can be regarded as 'hybridisation' rather than digitalisation. Hybridisation can help dismantle barriers and thereby reduce the digital literacy divide (Banerjee et al., 2023).

5.4. Critical juncture 4 – institutionalisation: DSI as an empowerment approach

All four cases resulted in long-term assignments of project structures, budgets, and responsibilities. However, these institutionalisations are based on different resources. Case C relies on exogenous resources that were acquired for further project development. By contrast, Cases A, B, and D identified endogenous resources for stable long-term project adoption. The basis of institutionalisation is the capacity building of administrative and voluntary partners throughout the process. Moreover, local political willingness was a central permissive condition in all cases. Those cases with bottom-up approaches (A, B) also found ways to re-activate public engagement and participation.

All four cases illustrate how a successful learning process uncovers potential within administrations, social organisations and voluntaries (Thewes et al., 2024). The institutionalisation processes in all cases can certainly be seen as evidence for project success. Moreover, they can be considered as evidence for triggering more general processes, namely learning, cooperation, and innovation. This confirms the strand of literature that sees DSI as empowerment (Anania and Passani, 2014). A lack thereof could also have directed the development processes in a different direction. Thereby, our study highlights the empowerment dimension within the DSI concept (Avelino et al., 2019; Bria et al., 2014; Rodrigo et al., 2019). As a result, rural digitalisation can develop effects beyond the project scope by establishing new capacities and regional assets.

6. Beyond digitalisation? Processes of rural (digital social) innovation

Our study demonstrates only a fraction of potential rural digitalisation trajectories, yet one with successful institutionalisation. While the findings may not be gerneralizable to all rural digitalisation projects, we still consider our findings plausibly representative for two types of projects, namely digitalisation and content approaches. Moreover, they provide a starting point for further quantitative research involving multiple cases. However, present the analyses already illustrate the potential of a critical junctures perspective that is backed by prominent literature in rural studies and DSI. First, we show that specific strategic decisions within digitalisation processes can be conceptualised as critical junctures in a comparable way across cases. Empirically, our case study suggests that these critical junctures neither have a conclusive character nor result in lock-ins, even if they might result from path-dependent decisions. Rather, critical junctures can be understood as windows of opportunity for change and innovation throughout processes (Soifer, 2012). The comparative approach highlights the variety of possible options in the decisions under examination. With regard to the first research question on the role of critical junctures in digitalisation, this underlines that each decision bears relevance on a more general level.

Second, different junctures trigger different elements of the SI/DSI concepts, including participation, process innovation, learning, and empowerment. The relevance of the dimensions of the DSI concept goes beyond the project scope. For example, rather than the actual spatial level of initial momentum, the differentiation regards a different aspect. Hence, underlying potentials and risks reveal much about their leverage to break path dependencies and create innovation in subsequent processes. Similarly, the purpose of participation may go beyond the involvement itself aiming to embed innovation in local structures and networks. Thus, the role of participation goes beyond empowerment and democratic access. In practice, this can also leverage long-term success by embedding digitalisation within social structures and networks and dismantle technological thresholds. Our comparative approach outlines digitalisation processes as a sequence of either endogenous or exogenous impulses. With this multi-level and temporally dynamic perspective, rural innovation integrates into a greater context, whereby innovation is seen to benefitfrom, but not be dependent on, exogenous factors. An analysis of rural innovation from a NERD perspective can, therefore, highlight some of its beneficial aspects, but as well emphasise process-related challenges.

Hence in response to our second research question, our case studies illustrate digitalisation as a driver of (digital) social innovation as such. Beyond this product innovation, digitalisation triggers new forms of cooperation and participation. Our case studies underline the potential of digital projects to empower local communities to experiment with new technologies and adopt this exogenous influence – whether intended or not. On this basis, our case studies provide insight into Digitalisation as a neo-endogenous stimulus to social innovation in rural areas. Digitalisation may empower rural communities to handle exogenous trends. In terms of practical rural development, the joint identification of problems and the search for solutions under the participation of various rural stakeholder groups can enhance local benefit and be an example of social innovation on a more general level.

In practice, locally embedded hybridisation can be an intermediate step to address the digital divide, including all related locational disadvantages and barriers. In this way, digitalisation fosters rural innovation in areas by creating opportunities for learning and collaborative capacity building, hence breaking path dependencies. Edogenously embedded digitalisation can increase social cohesion and place attachment within these local communities. Hybridisation can thus broaden the boundary conditions for further innovation.

Declaration of generative AI in scientific writing

Statement: During the preparation of this work the author(s) used DeepL & Grammarly in order to improve style and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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CRediT authorship contribution statement

Carola Sommer: Writing – Editing & Reviewing, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tobias Chilla:** Writing – Editing & Reviewing, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Lisa Birnbaum:** Writing – Editing & Reviewing, Investigation, Formal analysis, Data curation. **Stephan Kröner:** Writing – Editing & Reviewing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The paper relies on qualitative data. For data protection reasons, transcripts cannot be made available. Detail on documents used is provided.

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